

TOPICS

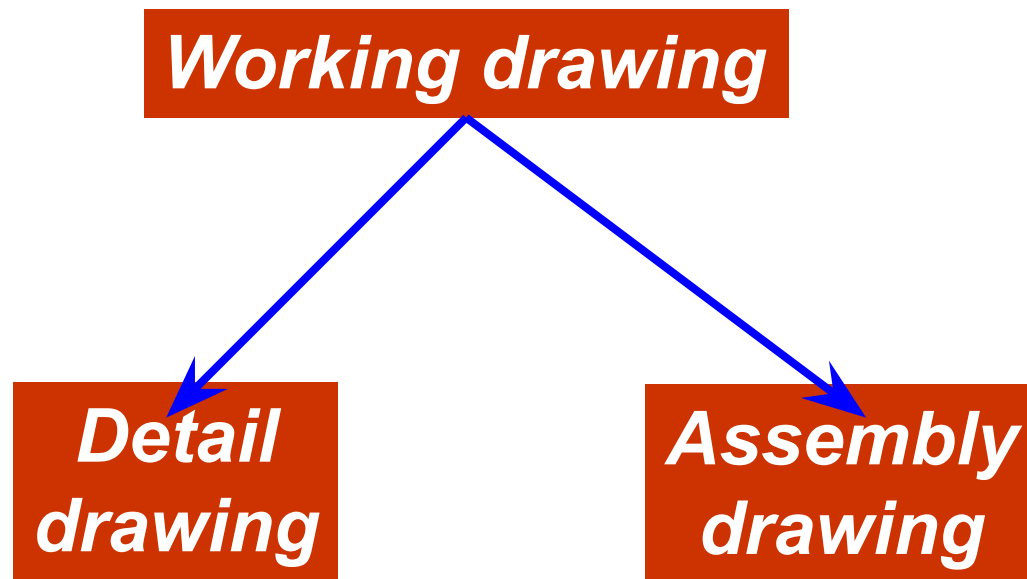
- Introduction
- Detail drawing
- Assembly drawing
- Assembly section
- Dimensioning

Introduction



DEFINITION

- ***Working drawing*** is a set of drawing used during the making/manufacturing of a product.



DEFINITION

- ***Detail drawing*** is a ***multiview representation*** of a single part with ***dimensions and notes***.

- ***Assembly drawing*** is a drawing of various parts of a machine or structure assembled in their relative working positions.

PURPOSE

■ *Detail drawing* conveys the *information* and *instructions* for manufacturing the part.

■ *Assembly drawing* conveys

1. completed shape of the product.
2. overall dimensions.
3. relative position of each part.
4. functional relationship among various components.

Assembly Drawing



TYPES OF ASSEMBLY DRAWING

1. Exploded assembly drawings

The parts are separately display, but they are aligned according to their assembly positions and sequences.

2. General assembly drawings.

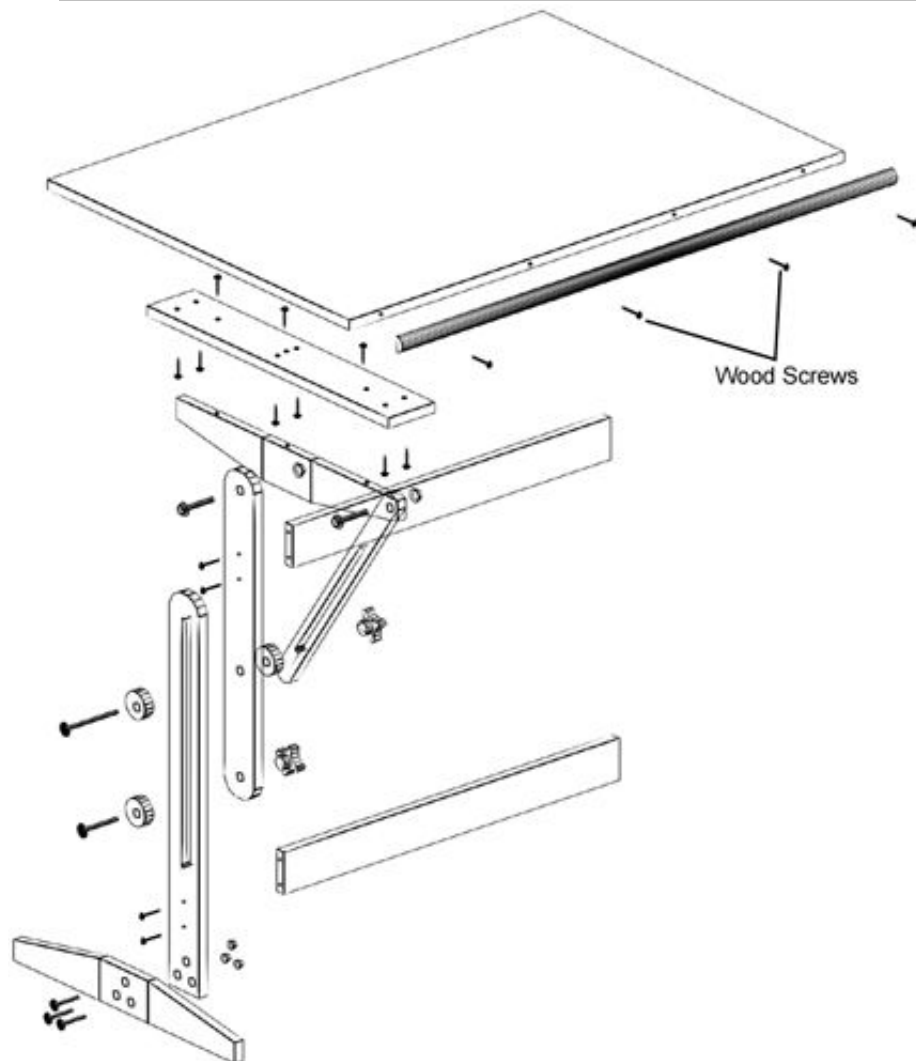
All parts are drawn in their working position.

3. Detail assembly drawings

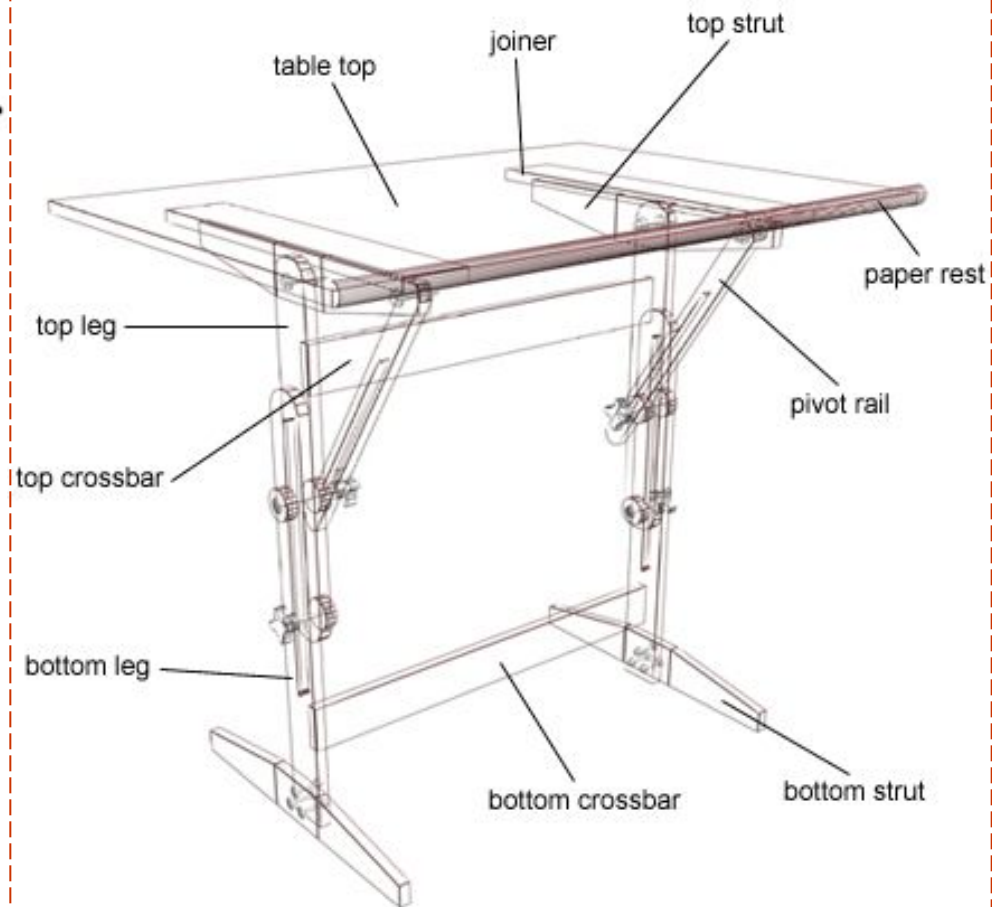
All parts are drawn in their working position with a completed dimensions.

1. EXPLODED ASSEMBLY

Pictorial representation

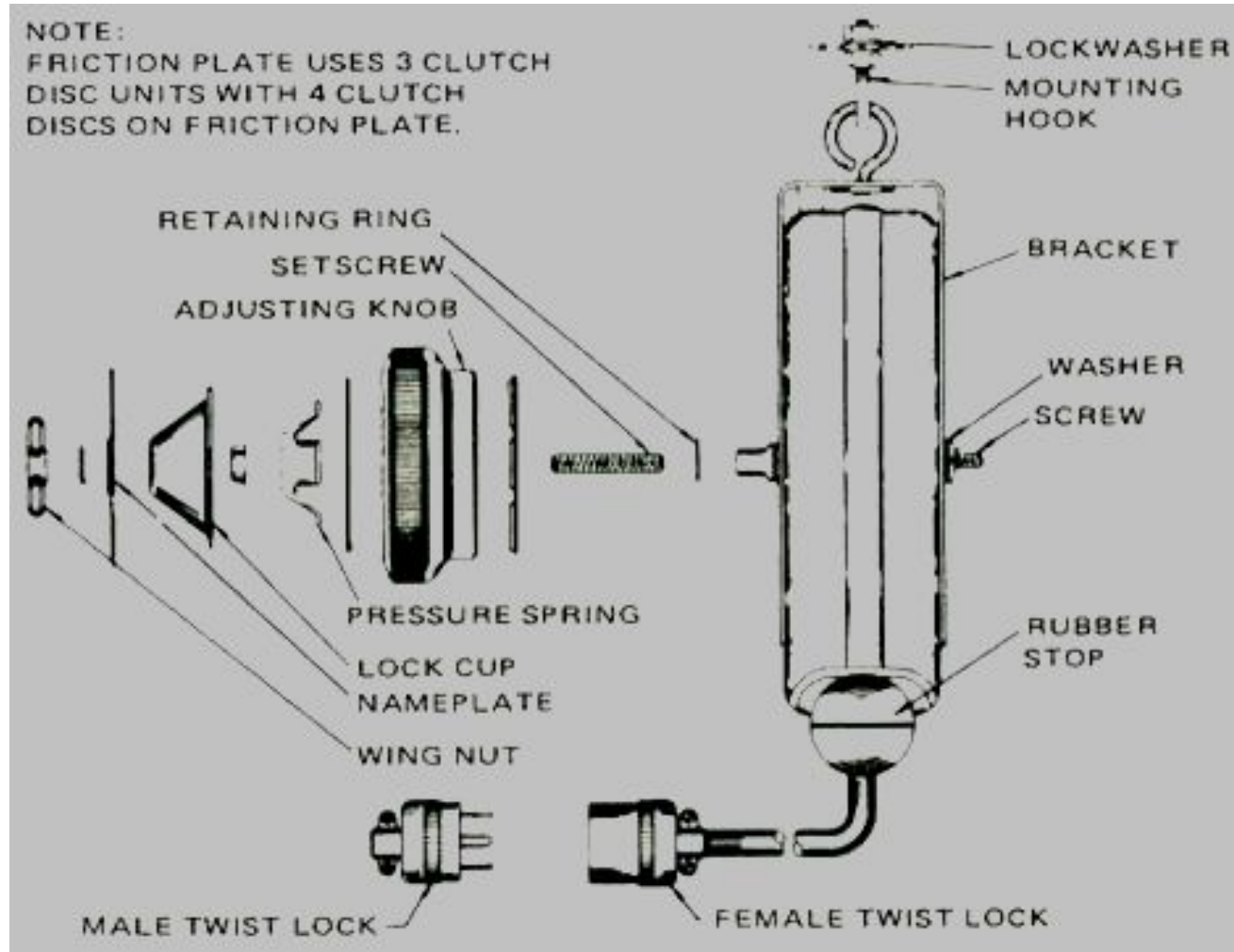


Finished product



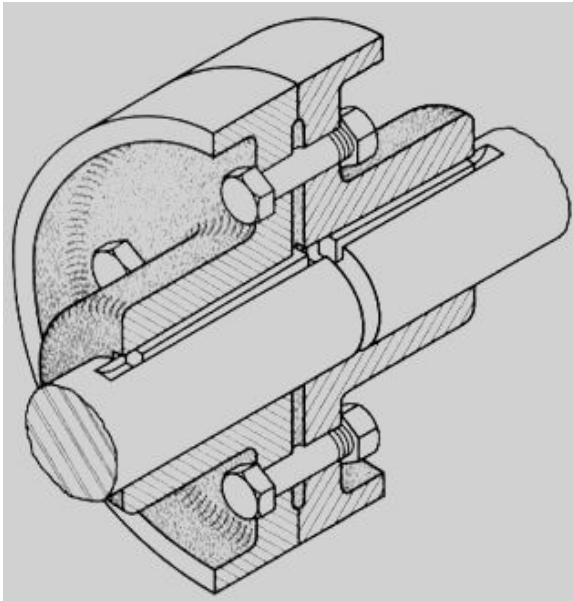
1. EXPLODED ASSEMBLY

Orthographic representation

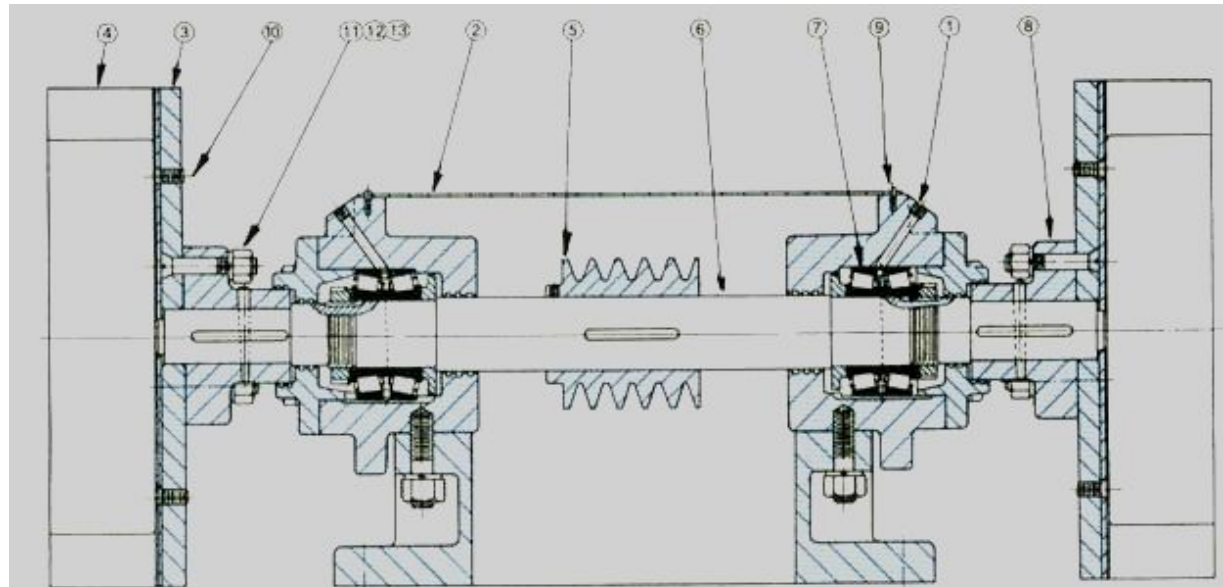


2. GENERAL ASSEMBLY

Pictorial

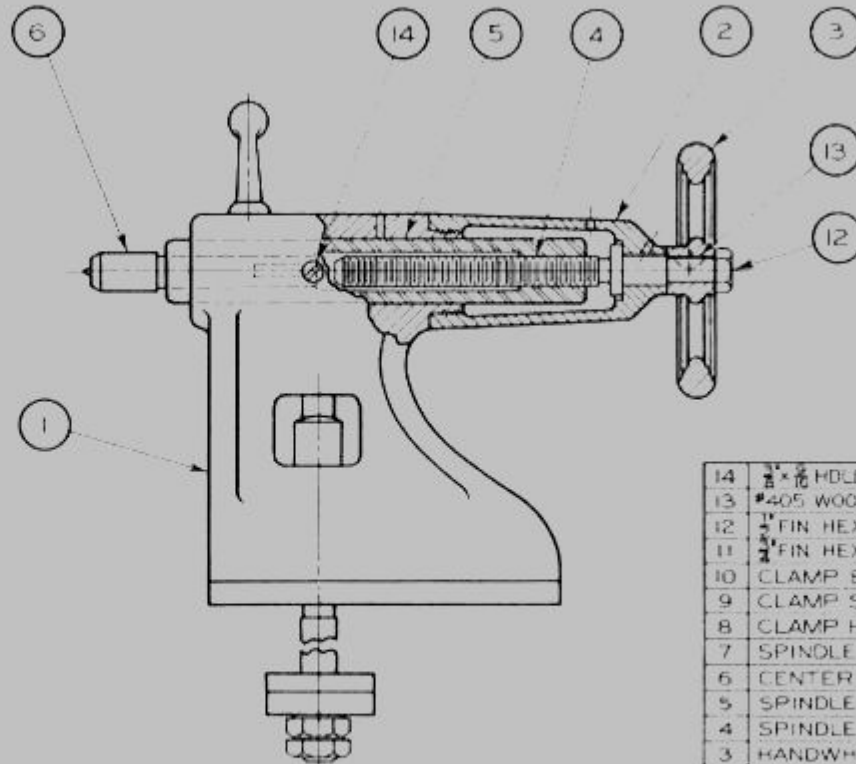
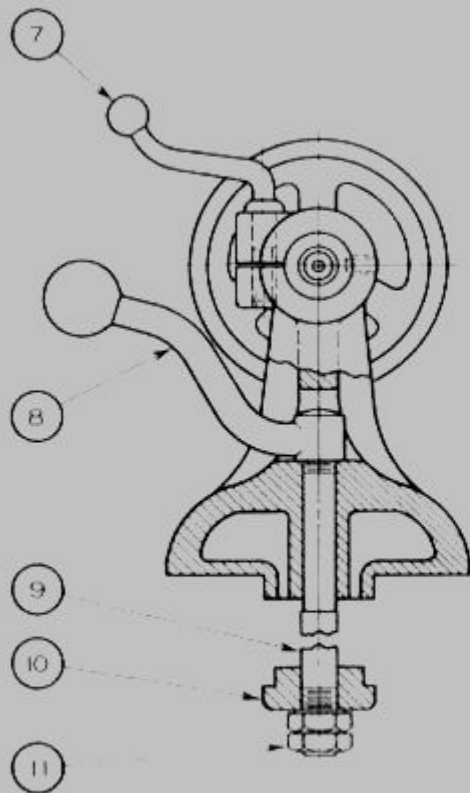


Orthographic



2. GENERAL ASSEMBLY

10D-001



14	3/8" x 16" HDLESS SET SCR. FL DOG PT	1	
13	#405 WOODRUFF KEY	1	
12	1/2" FIN HEX NUT	1	
11	3/4" FIN HEX JAM NUT	2	
10	CLAMP BLOCK	SAE III2	1
9	CLAMP STUD	SAE III2	1
8	CLAMP HANDLE	SAE III2	1
7	SPINDLE HANDLE	SAE III2	1
6	CENTER	SAE I315	1
5	SPINDLE	SAE III2	1
4	SPINDLE SCREW	SAE III2	1
3	HANDWHEEL	C I	1
2	SPINDLE SCR BRG	C I	1
1	TAILSTOCK BASE	C I	1
NO	PART NAME	MAT'L	REQD

TITLE ASSEMBLY OF TAILSTOCK

MACHINE 10D SHOP ORDER 1064 DATE 3-20-54

DRAWN BY JRS CHECKED H R APPROVED LMT

MATERIAL HARDNESS

HEAT TREATMENT

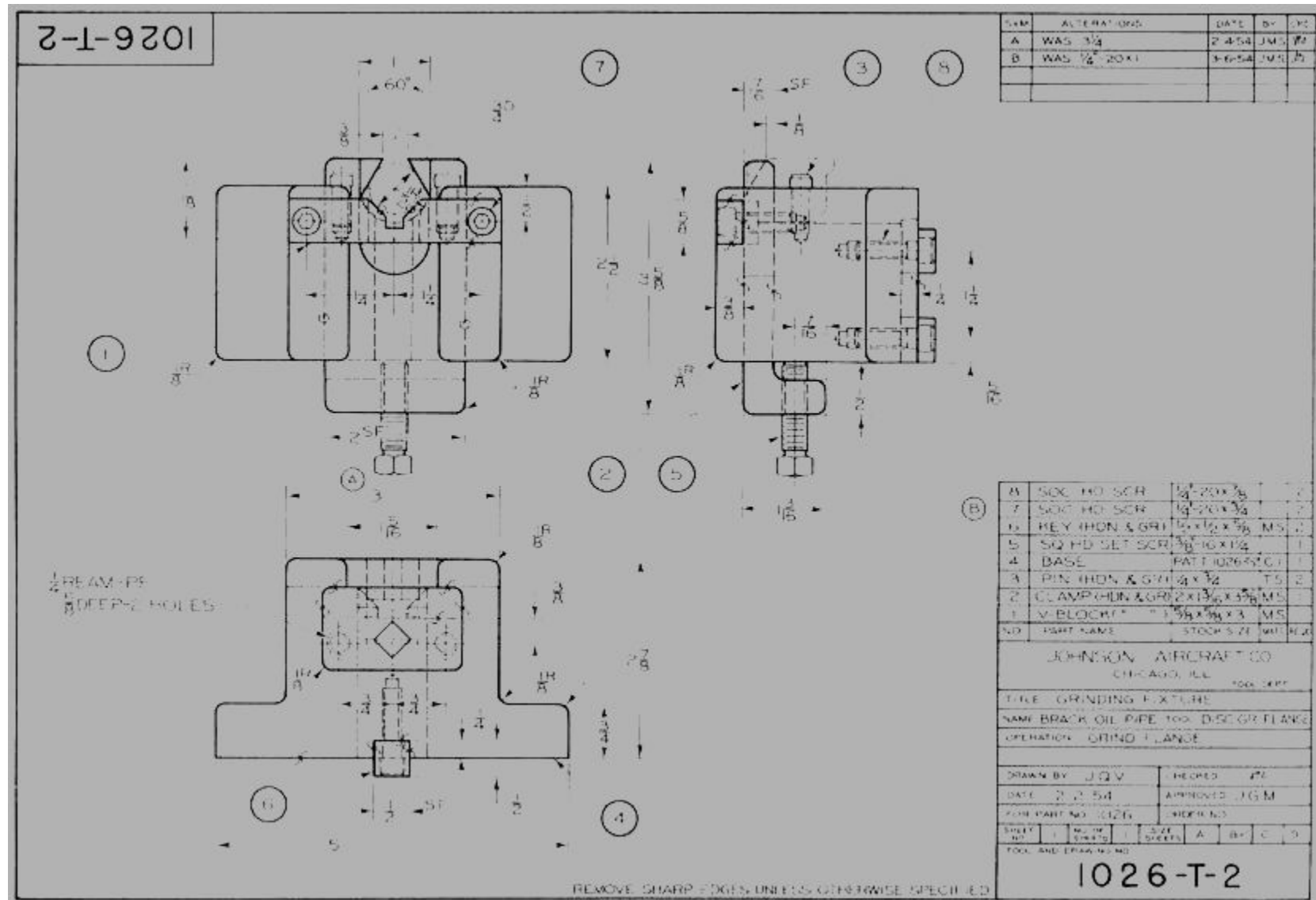
JACKSON MACHINE CO
ST LOUIS, MO

10D-603

LETTER CHANGE BY DATE

3. DETAILED ASSEMBLY

(working-drawing assembly)

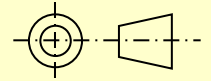


REQUIRED INFORMATION IN GENERAL ASSEMBLY DRAWING

- 1. All parts, drawn in their operating position.**
- 2. Part list (or bill of materials, BOM)**
 1. Item number
 2. Descriptive name
 3. Material, MATL.
 4. Quantity required (per a unit of machine), QTY.
- 3. Leader lines with balloons around part numbers.**
- 4. Machining and assembly operations and critical dimensions** related to operation of the machine.

PLACING AN INFORMATION

(This course)



- Assembled parts
- Reference numbers

General notes

Part list

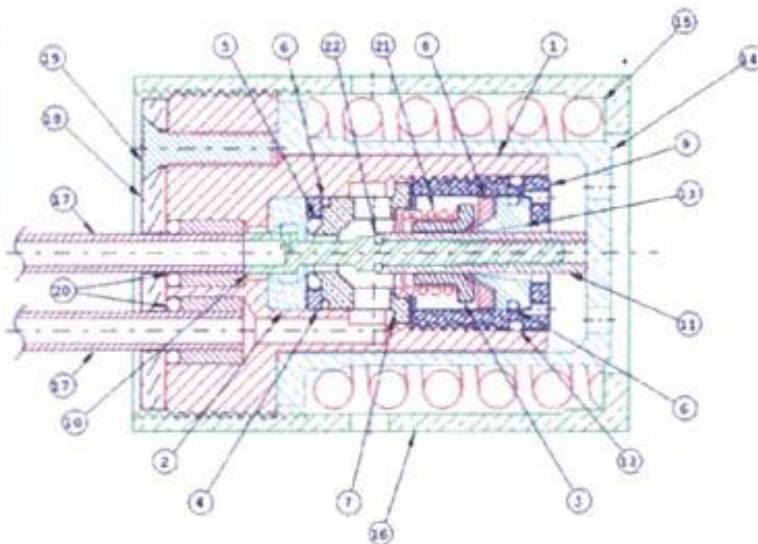
Title block

PART LIST (BOM) (This course)

- Locate above the title block.
- Fill the table from the bottom.

3	SET SCREW	1	Stainless Steel, M3 HEX SOCK CUP PT
2	SHAFT	1	Stainless Steel
1	SUPPORT	2	Cast Iron
NO.	PART NAME	REQD.	MATL. & NOTE

EXAMPLE : Another allowable place for BOM



BILL OF MATERIAL				
FIND NO.	PART NO.	DESCRIPTION/TITLE	NO. REQD	REMARKS
1	BE2001	BODY	1	
2	BE2002	SEAL SUPPORT	1	
3	BE2003	SEAL	1	
4	BE2004	SPACER RING	1	
5		O-RING .100 I.D. x .040 W	1	
6		O-RING .220 I.D. x .032 W	2	
7	BE2005	MANIFOLD	1	
8	BE2006	TENSION SEAL	1	
9	BE2007	RETAINING NUT	1	
10	BE2008	POPPET STEM	1	
11	BE2009	POPPET SEALING SLEEVE	1	
12		O-RING .055 I.D. x .032 W	1	
13	BE2010	SEAL SUPPORT	1	
14	BE2011	CAGE 1	1	
15		SPRING	1	
16	BE2012	CAGE 2	1	
17	BE2013	PORT TUBE	2	
18	BE2014	CLAMPING DISC	1	
19		HOLD DOWN SCREW	3	
20		O-RING .097 I.D. x .035 W	2	
21		ENERGIZER SPRING	1	
22		O-RING	1	

 STONE ENGINEERING COMPANY A DIVISION OF THE SPAS CORPORATION 305 MADISON ST., SUITE 2C MONTVILLE, AL. 35461			
DRAWN BY: TLA DATE: 09/21/93 SCALE: 3:1	APP. BY: DATE: 09/21/93 SCALE: 3:1	TOLERANCES: FRACTIONS DECIMALS .015 ± .001 .001 ± .001	
TITLE: HYDRAULIC REGULATOR ASSEMBLY		DRAWING NO.: BE1001	

REVISED	DESCRIPTION	DATE	BY	APPROV

STEPS TO CREATE ASSEMBLY DRAWING

1. **Analyze** geometry and dimensions of all parts in order to understand the *assembly steps* and overall shape of device or machine.
2. Select an appropriate view.
3. **Choose major parts**, i.e. parts that have several parts assembled on.
4. Draw a view of **major parts** according to a selected viewing direction.

STEPS TO CREATE ASSEMBLY DRAWING

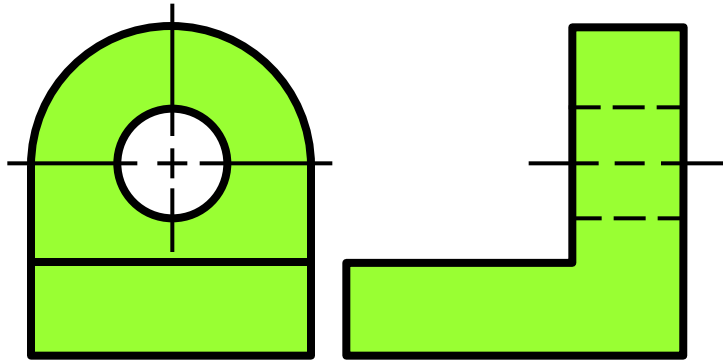
5. Add detail view of the remaining parts at their working positions.
6. Apply ***section technique*** where relative positions between adjacent parts are needed to clarify.
7. Add ***balloons***, ***notes*** and ***dimensions*** (if any).
8. Create BOM.

GENERAL PRACTICE

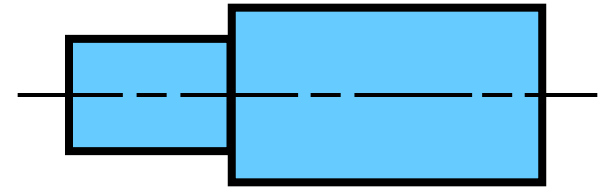
- The *number of views* can be one, two, three or more as needed, but it should be **minimum**.
- A good *viewing direction* is that represents all (or most) of the parts assembled in their working position.

EXAMPLE : Selection of a necessary view

Given

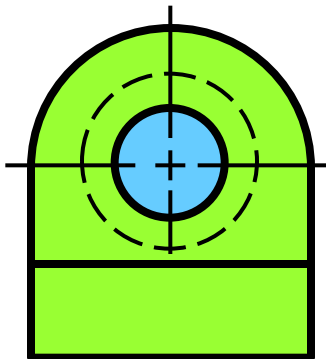


Part A

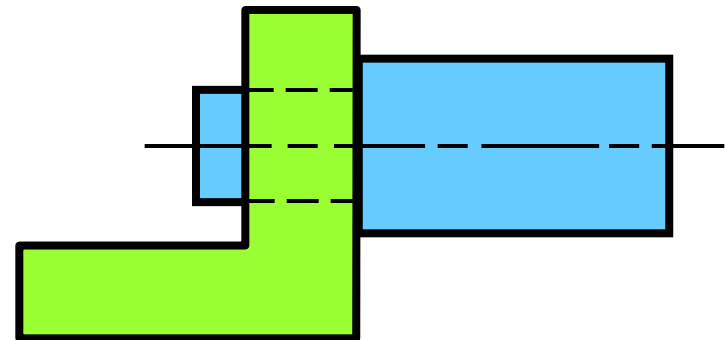


Part B

Student A



Student B

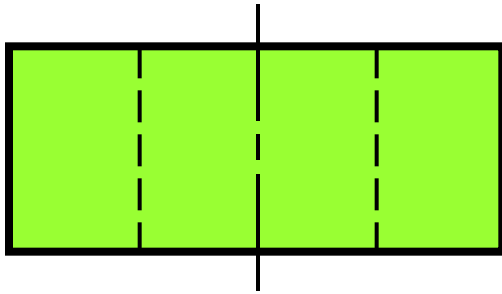


Which is an appropriate view for assembly drawing ?

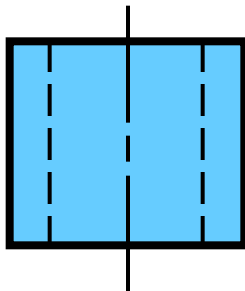
GENERAL PRACTICE

- *Hidden lines* usually **omit** unless they are absolutely necessary to illustrate some important feature that the reader might otherwise miss.

EXAMPLE : Hidden lines omit *or not* ?

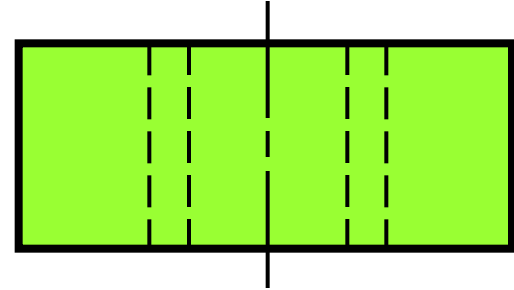


Part A

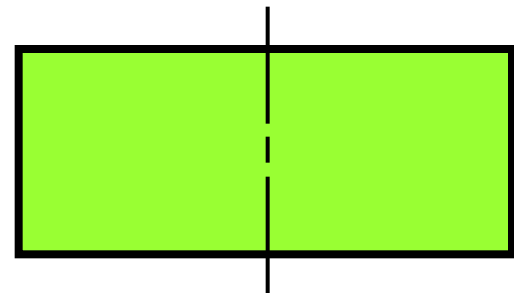


Part B

A



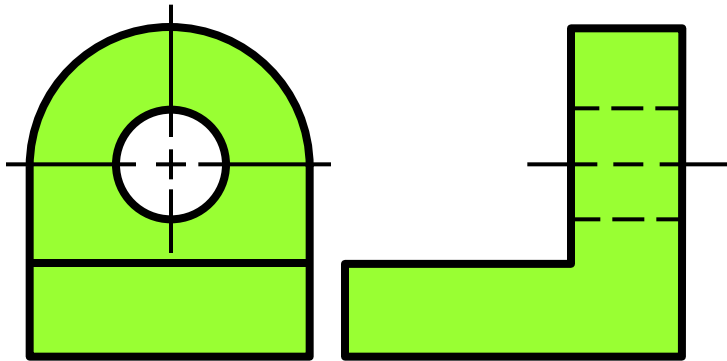
B



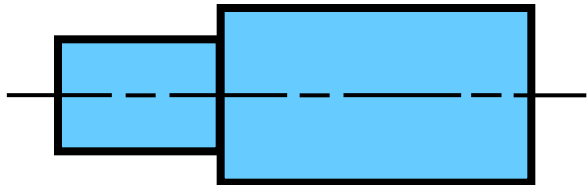
Good

Poor

EXAMPLE : Hidden lines omit *or not* ?

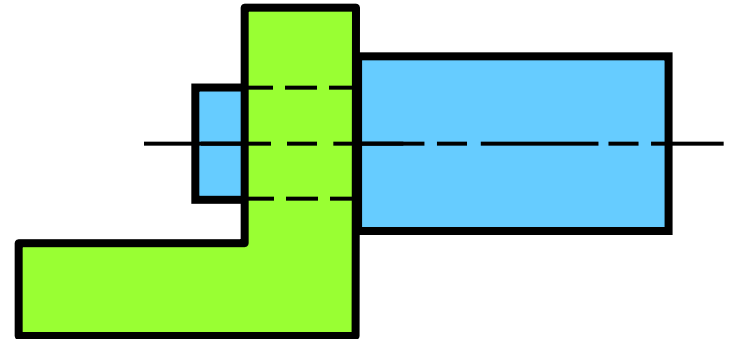


Part A

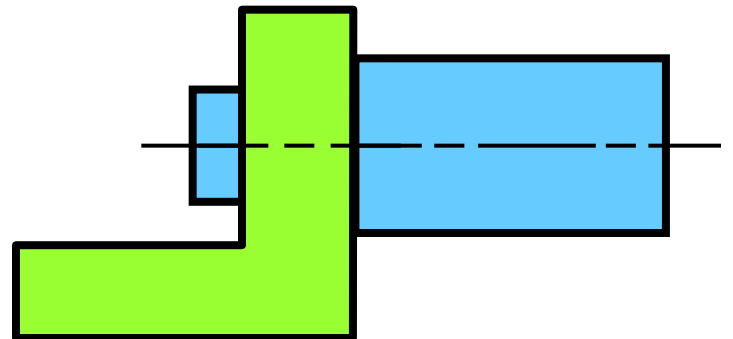


Part B

A



B



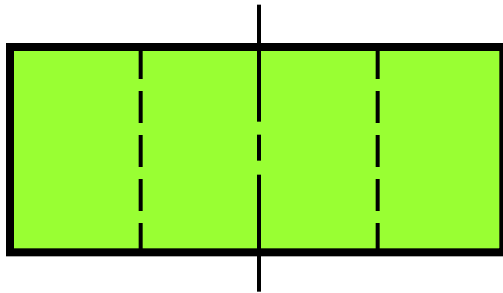
Good

Poor

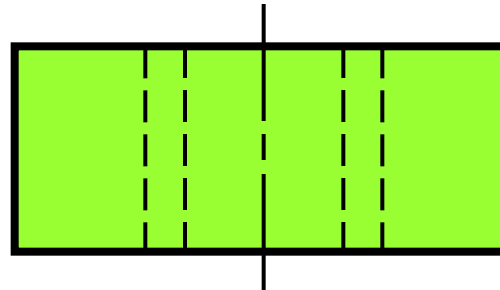
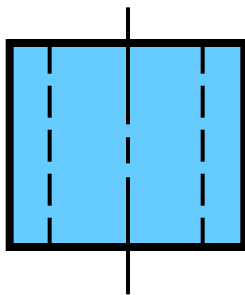
GENERAL PRACTICE

- ***Section technique*** is usually need to clarify mating of the parts.
- Use ***different*** section line styles for adjacent parts.

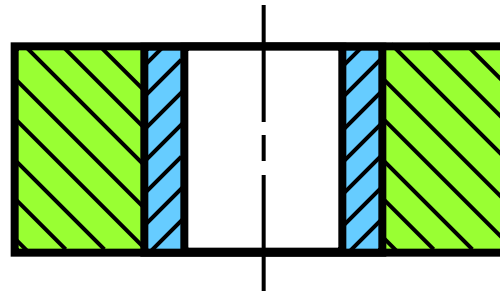
Part A



Part B



Correct

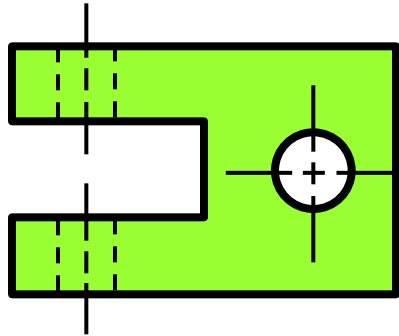


Better

SECTION LINE PRACTICE

- ***Do not*** draw section lines on sectional view of standard parts.
 - Threaded fastener
 - Washer
 - (longitudinal cut of) Solid shaft, Pin, Key

EXAMPLE 1 : Assembly steps



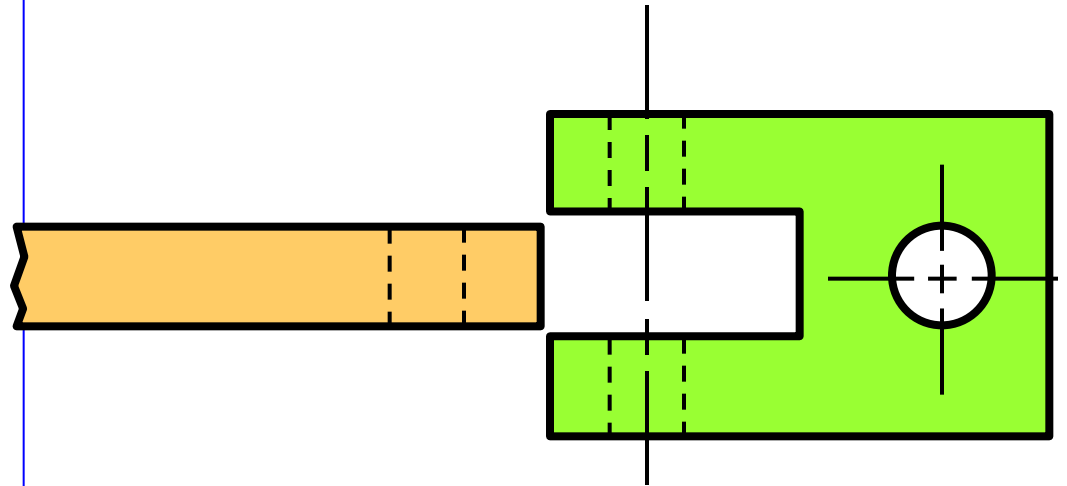
① CLEVIS, Steel, 1 REQD.



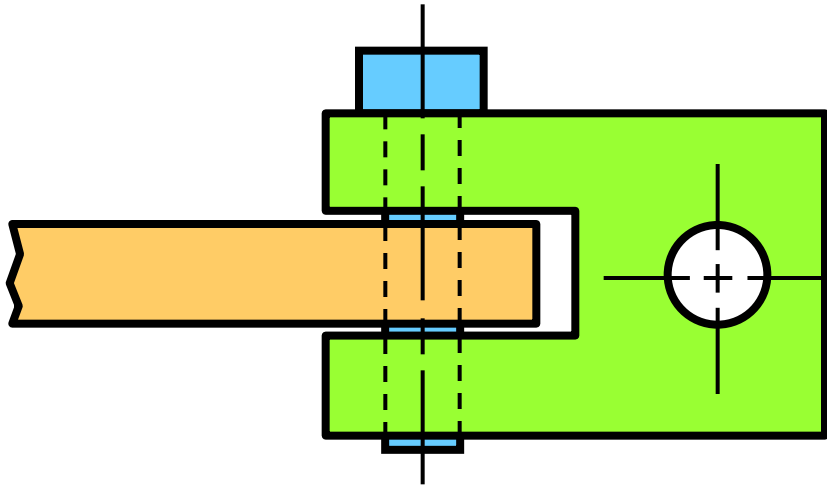
② ARM, Steel, 1 REQD.



③ PIN, Steel, 1 REQD.



EXAMPLE : Section line practice



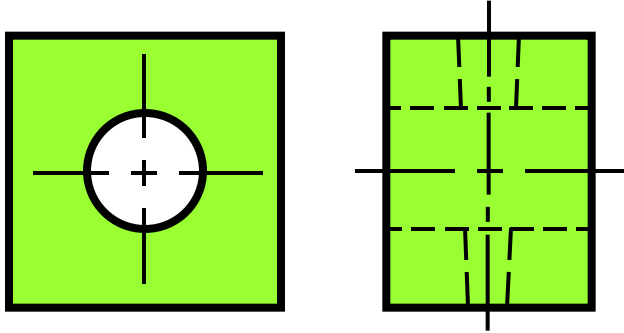
Which is an appropriate full section view of this assembly ?



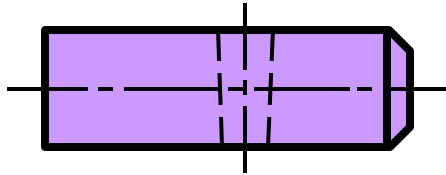
Good

Poor

EXAMPLE 2 : Assembly steps



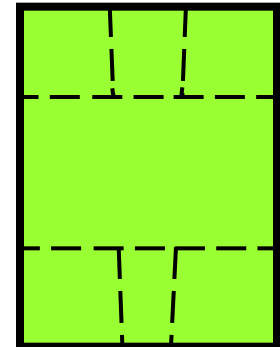
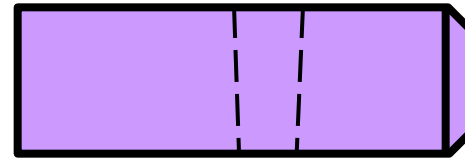
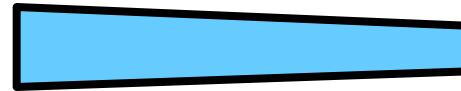
① SUPPORT, Steel, 1 REQD.



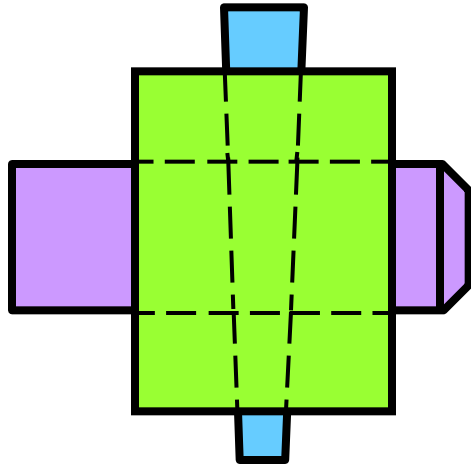
② SHAFT, Steel, 1 REQD.



③ TAPER PIN, Steel, 1 REQD.

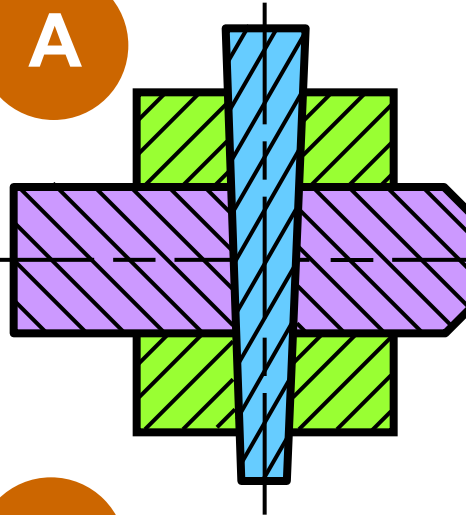


EXAMPLE : Section line practice

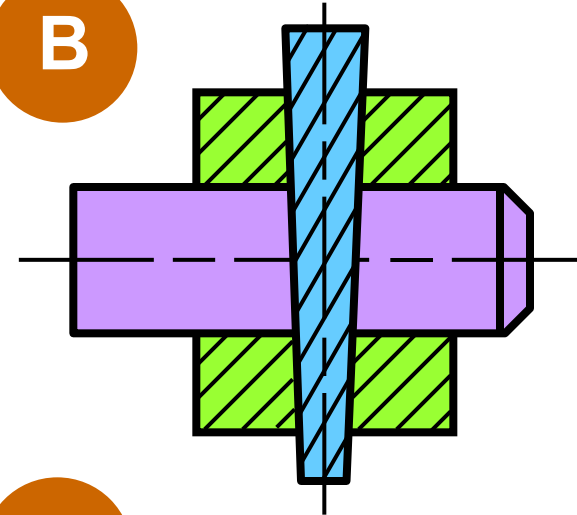


Which is an appropriate full section view of this assembly ?

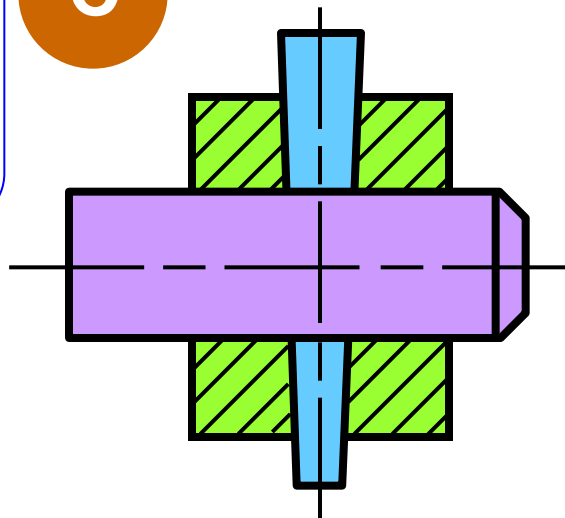
A



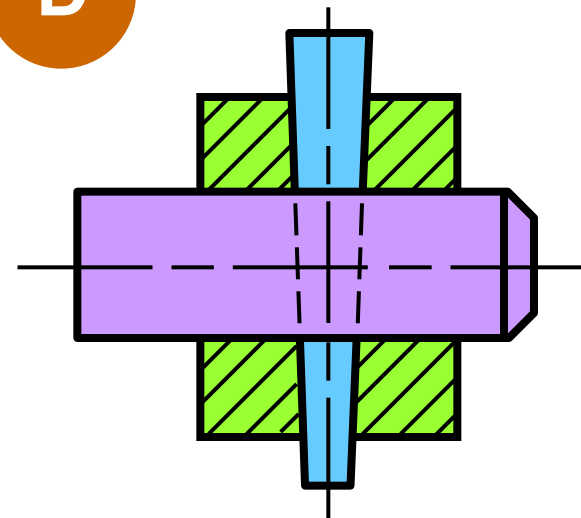
B



C



D

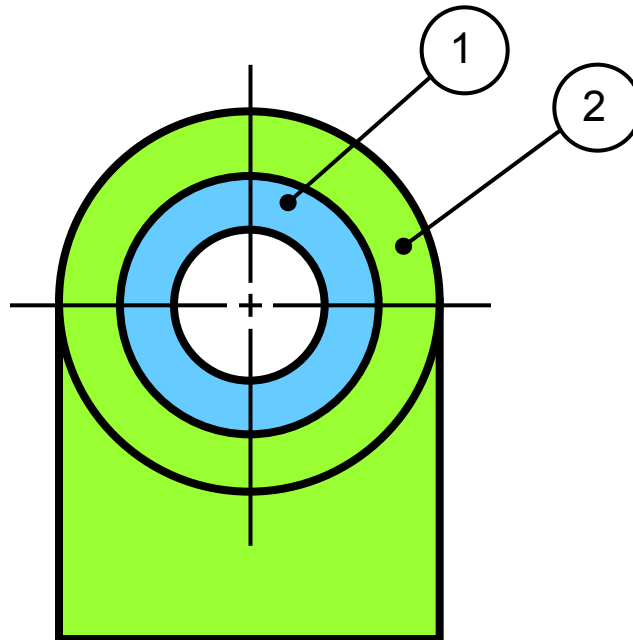


Good

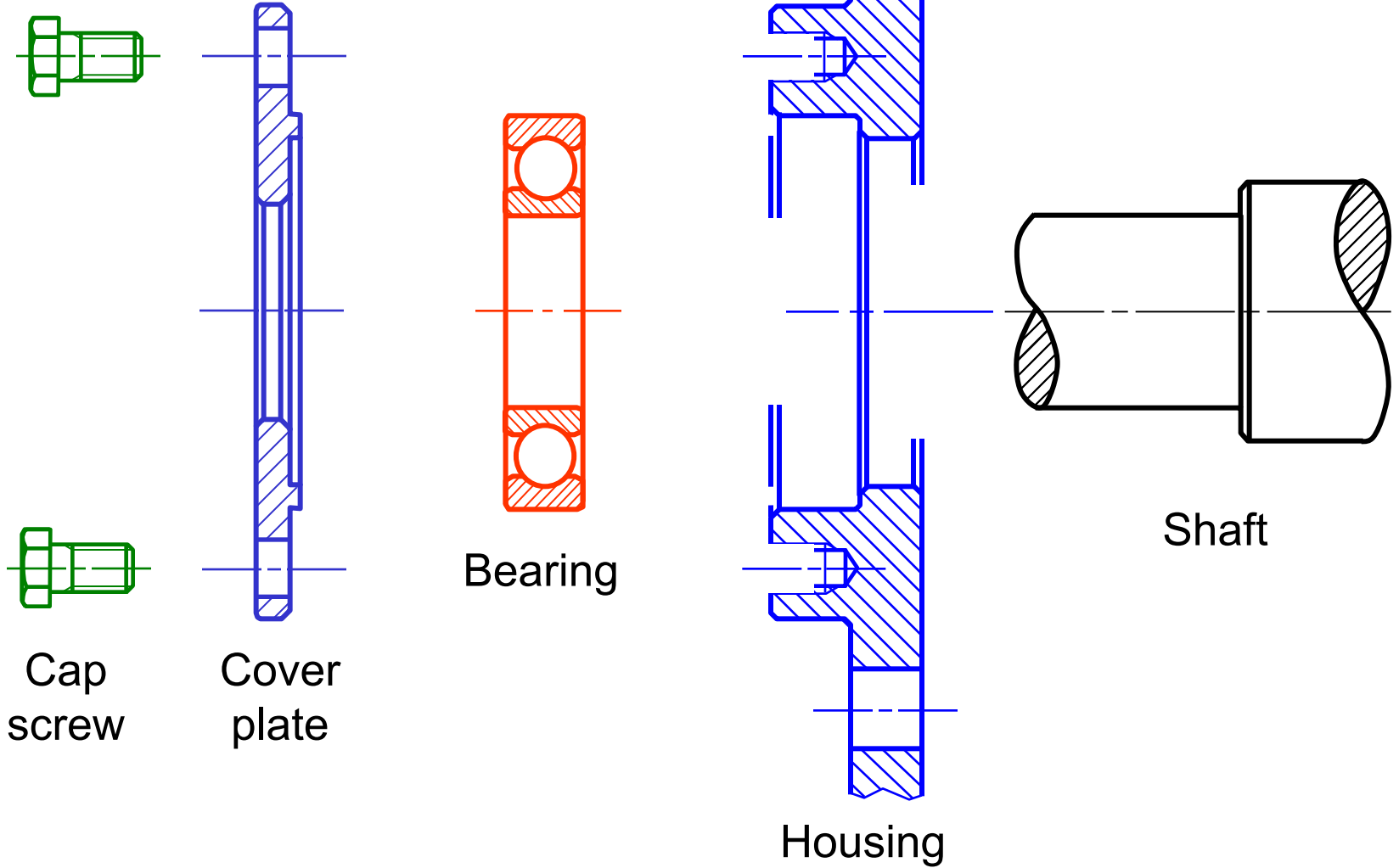
Poor

LEADER LINE PRACTICE

- Drawn from the inside of the part to the balloon and placed a filled circle at the beginning of a line.
- Drawn in the *oblique* direction.



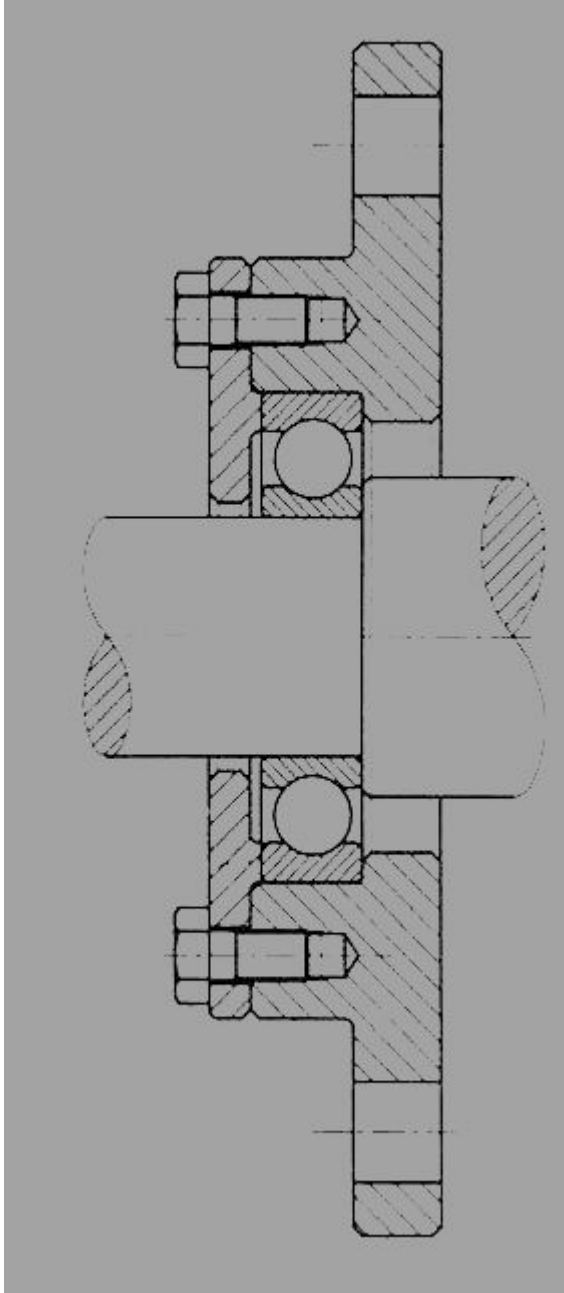
EXAMPLE



INTERPRETING ASSEMBLY DRAWING

1. Assemble steps.
2. Function of each part in machine.
3. Design concept.

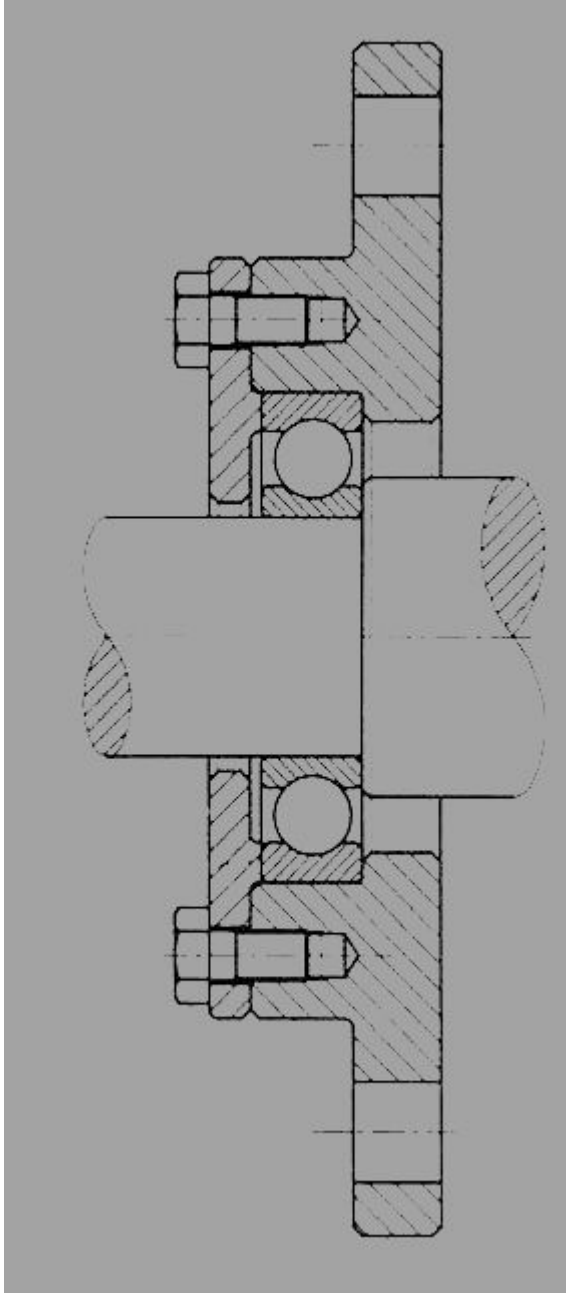
EXAMPLE 1 : *Shaft support on a machine housing*



Assemble steps

1. Install bearing to the shaft.
2. Install the bearing-shaft unit to the housing.
3. Install the cover plate.
4. Tighten the screw.

EXAMPLE 1 : *Shaft support on a machine housing*



Functions of main parts

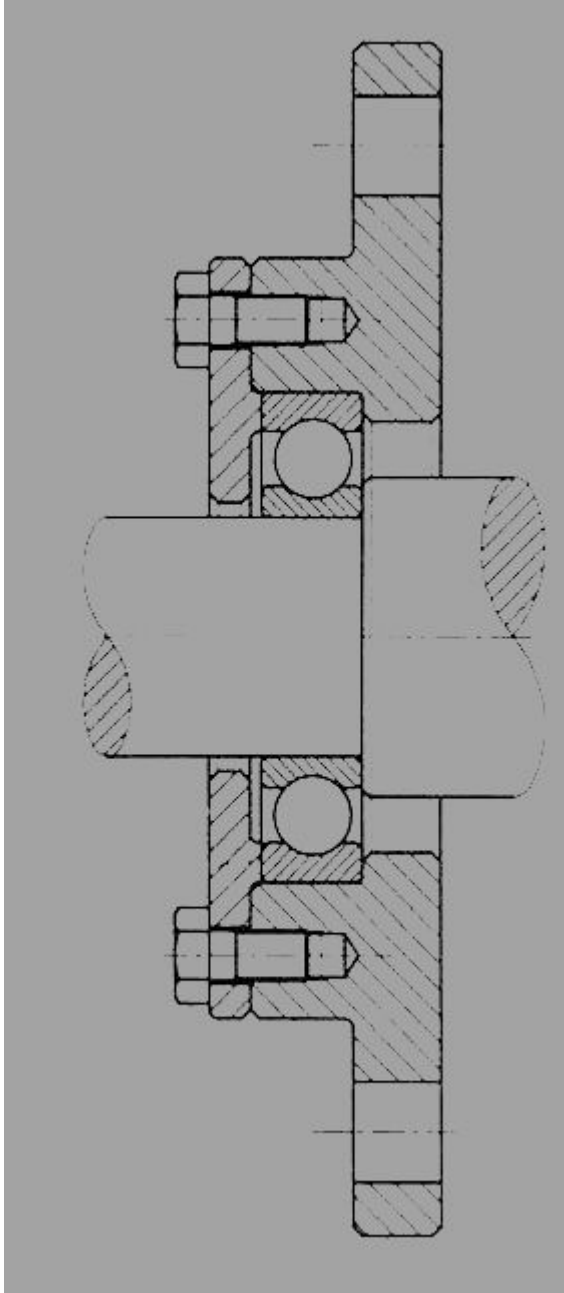
1. *Bearing* :

Support the rotating shaft.

2. *Cover* :

- Control an axial movement.
- Prevent the bearing unit from rotation.

EXAMPLE 1 : *Shaft support on a machine housing*



Design concept

Avoid direct contact between rotating shaft and housing as well as cover plate by using a bearing and clearance holes.

Mating of Parts



POINTS TO CONSIDER

1. Surface finishing

2. Tolerance

- Size
- Geometry

SURFACE FINISHING

- Surface finishing means the quality of a surface. It relates to the level of roughness of a surface.

Purpose

1. To control the accuracy in positioning and tightness between mating parts.
2. To reduce the friction, especially for the part moves relative to other parts.

TOLERANCE

- Tolerance is the total amount dimension may vary.
It is defined as the difference between the upper and lower limits.

Purpose

1. To control an ***interchangeability*** of parts.
2. To ensures the mating part will have a desired fit.